Prosenjit Mondol (Prosen)

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EDUCATION

Patuakhali Science & Technology University (PSTU) **February 2023 – Present**
B.Sc. in Computer Science and Engineering (Dean's List) Cumulative GPA: 3.69/4.00 (Credits Completed 126.5 out of 165)

WORK EXPERIENCE

Research Software Engineer & Lab Systems Coordinator **August 2025 – Present**
Computation Informatics & Machine Intelligence Lab **(CIMILab)** University of Missouri, USA

RESEARCH INTERESTS

AI in Health, Health Informatics, Human-Computer Interaction, Machine Learning, Image Processing, Computer Vision

PUBLICATIONS

Accepted: " A Sensitivity-Gated Cascade for Compute-Efficient Spinal Lesion Detection in Resource-Constrained Settings" Accepted for presentation at the 3rd Medical Image Computing in Resource-Constrained Settings Workshop & Knowledge Interchange (MIRASOL), MICCAI 2026, Strasbourg, France. ***First Author(2026)***

Presented: "A Deep Learning Approach to Bangla License Plate Text Extraction using Easy OCR." Presented at International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure 2026. [Certificate]

In Progress: "DSANet: A Grid-Search Optimized Heterogeneous Ensemble for Ischemic Stroke Lesion Segmentation" Submitted at 14th International Conference on Electrical and Computer Engineering, BUET, Dhaka, Bangladesh. ***First Author(2026)***

SELECTED AWARDS & HONOURS

PSTU Dean's Merit Award: Awarded scholarships for academic excellence among 84 students.

Aungkur Scholarship: Provided by Patuakhali Science & Technology University.

PSTU Programming Contest: Ranked in the top 5 out of 50 teams in 2025.

Merit Scholarships: Government Board Merit scholarships for outstanding results in both Higher Secondary Certificate (2021) & Secondary School Certificate (2019).

TECHNICAL STRENGTHS

Programming Languages	R, Python, C++, Java, JavaScript, HTML/CSS, PHP, C#, C, dart
ML Framework / Library	Scikit-learn, PyTorch, Pandas, Numpy, Keras, TensorFlow, OpenCV
Data Visualization/Others	Matlotlib, Seaborn, Plotly
Database	MySQL, Oracle, SQL Server
Backend	Laravel, Flutter, Dotnet
Miscellaneous Tools	Git, Latex, Jupyter Notebook, VS Code, Android Studio

COMPETITIVE PROGRAMMING

CodeForces - prosenjit_mondol **HackerRank -** prosenjit1156 **LeetCode -** prosenjit_mondol **Beecrowd -** Prosenjit

- Solved 1500+ problems across multiple competitive programming platforms, developing expertise in algorithms, data structures, and problem-solving techniques.
- Actively competed in prominent programming contests, including the ICPC Dhaka Regional Preliminary (2023–2025), BitFest KUET (2024), and the PSTU IT Carnival.

PROJECTS

NeuroVision AI
Git Repository OR **View Live**

NeuroVision AI is a cutting-edge medical imaging platform designed to assist doctors, radiologists, and researchers in diagnosing neurological and cardiovascular conditions through advanced deep learning models. The platform integrates multiple AI pipelines for brain, spine, heart, and chest imaging, providing lesion detection, segmentation, explainable AI (Grad-CAM), and research monitoring in a secure, enterprise-grade environment.

TrashBot
Git Repository OR **View Live**

Developed TrashBot, an AI-powered smart waste management system using Raspberry Pi 3 Model B, computer vision, and IoT technologies. The project integrated YOLO-based real-time waste detection, deep learning-based classification, live web monitoring, and remote tracking features to support intelligent waste collection and segregation for smart city applications.

Graph-Visualizations
Git Repository OR **View Live**

Graph Craft is a web-based tool for visualizing and analyzing graphs. Provides an interactive environment where users can input their graph data in the form of an adjacency matrix and instantly visualize the graph along with various analysis results.

Reuse Hub
Git RepositoryOR View Live

Developed and deployed a production-ready, community-driven platform using Flutter 3.x (Dart) and the Firebase Ecosystem (Firestore, Auth, Storage). The primary objective of ReuseHub is to reduce waste and promote sustainability by connecting donors with seekers of reusable goods, facilitating a circular economy

Auto Checking BOT
Git RepositoryOR View Live

Designed and developed an automated credential verification bot using Python. The bot reads email-password pairs from an Excel sheet, attempts authentication on a target website, and logs valid credentials into a separate Excel file. Built with automation logic to run continuously until all entries are processed, significantly reducing manual effort in large-scale validation tasks.

Smart Vacuum Cleaner
Git Repository OR View Live

The application is used to initiate the robot. The navigation of the robot is according to the help of sensors it detects and avoids obstacles. To save the time of the people the smart vacuum cleaner helps to clean the surface of the floor without any human intervention.

JOB PORTAL
Git Repository

Developed a dynamic and user-friendly Job Portal to streamline the recruitment process by connecting job seekers with employers. Implemented core features such as job posting, user authentication, application management, and advanced job search functionality using PHP, MySQL, HTML, JavaScript, and CSS.

Relevant Coursework of Undergraduate Program

Computer Science: Programming Language, Data Structure and Algorithms, Software Engineering, Database Systems, Numerical Methods, Discrete Mathematics, Artificial Intelligence, VLSI, Algorithm Engineering, Compiler Design and Automata Theory, Data Warehousing and Mining

Signal Communication: Data Communication Engineering, Optical Fiber Communication, Computer Networks, Network Routing and Switching and related lab courses.

Mathematics: Differential and Integral Calculus, Co-ordinate Geometry and Linear Algebra, Statistics, Vector Analysis and Complex Variables, Ordinary and Partial Differential Equations.

Electrical & Electronics: Basic Electrical Engineering, Electrical Technology, Digital Logic Design, Microprocessor and Interfacing and related lab courses.